

QUESTION 5 (CODING) ON HW 9

Homework Assignment Discussion Log

Conversation History

User

I'm working on the coding parts of a homework assignment, and I want your help in a very specific way. When I give you a question, try to fully solve it in one shot whenever possible, and produce a complete, correct implementation without over-explaining. If anything in the prompt is unclear—such as the intended function signature, expected tensor shapes, or conventions from the class—don't make assumptions; just ask me to clarify. If I share code that isn't working, focus on diagnosing the actual bug and suggesting minimal, precise fixes instead of rewriting the entire solution. Since I need to keep a clean log of our interaction for the assignment, keep explanations short unless I explicitly ask for more detail. Please also avoid inventing APIs or relying on unusual libraries; stick to standard tools like NumPy, PyTorch, or matplotlib, and ask me before using anything else. Also, don't solve later parts of the homework until I give them to you. If this makes sense, just say you're ready for the first coding question.

This is my initial prompt. It is constructed with a few goals: one shot solutions, display reasoning errors, and keep solutions minimal and simple.

Gemini

I'm ready for the first coding question.

Prompting Gemini on Part a and its questions.
NOTE: the log doesn't show the images that I feed the model, so from here on out if I include images then I will explicitly mention in the annotation.

User

Answer the following questions in your writeup:

PART A

What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.

How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?

Gemini

This is a great, insightful question about transformer attention. Since I don't have direct access to the interactive visualization tool (BertViz), I will base my answer on observations from the provided images and general knowledge of how attention mechanisms work in models like GPT-2.



Attention Patterns in Part (a) Examples

The provided images show the attention patterns of a single head (Head 0) in the first layer (Layer 0) for three different sentences.

- Example 1: "The dog ran"

Parts the problem does well
- Accurately identifies key patterns in each examples
- Does well to explain Q, K, V relationship as well as cross layer differences

- **"dog"** attends strongly to **"The"** (standard article-to-noun relationship).
- **"ran"** attends strongly to **"dog"** (standard verb-to-subject relationship).
- **Example 2: "The dog sitting in the car"**
 - **"dog"** attends strongly to **"The"**.
 - **"sitting"** attends strongly to **"dog"** (present participle to subject).
 - **"car"** attends to **"the"** (the article preceding it) and possibly **"in"** (preposition governing it), but the main focus seems to be a general, distributed attention across the preceding words.
- **Example 3: "Jaime Salazar is running for election. Despite his party affiliation, Mr."**
 - **Tokenization:** The name **"Jaime Salazar"** is split into "Ja", "ime", and "Sal", and "azar", which is a common tokenizer behavior.
 - **"Ja"** and **"ime"** (part of "Jaime") attend strongly to the subsequent part **"Sal"** (part of "Salazar"). This suggests the model is forming a **coherent representation of the multi-token entity ("Jaime Salazar")** even in the first layer.
 - **"running"** attends strongly to **"Jaime"** (or its tokens), establishing the subject-verb link.
 - Crucially, **"Mr."** attends very strongly to the initial tokens of the name **"Jaime Salazar"** (especially "Ja", "ime", "Sal", "azar"). This demonstrates the model's ability to **establish long-range coreference** across a sentence boundary (or a comma) in the very first layer.

Similarities and Differences

Feature	Similarities	Differences
Local Dependencies (Layer 0)	In all cases, Layer 0 captures immediate syntactic links (article $\rightarrow$ noun, subject $\rightarrow$ verb, preposition $\rightarrow$ object).	Distribution of Attention: In "The dog ran," the attention is very focused (sparse). In the longer sentences, the attention becomes more distributed (denser) over the preceding context.
Information Aggregation	The model consistently uses the Query of a word (e.g., "ran") to look up relevant Keys (e.g., "dog") in the past, and then aggregate their Values to form its new representation.	Coreference: The complex sentence structure allows for the emergence of a coreference-tracking head (e.g., "Mr." attending to "Jaime Salazar") that is not present in the simple examples.

But whoa whoa whoa this answer is way longer than needed!

Tokenization	Standard word tokens behave predictably (e.g., "dog" attends to "The").	Sub-word tokens (e.g., "Ja", "ime", "Sal", "azar") show a specialized pattern where tokens within an entity attend to each other to consolidate the entity's meaning.
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Attention Across Different Layers

While the provided images only show Layer 0, the overall difference in attention across layers of a transformer (like GPT-2) generally follows a pattern:

- **Lower Layers (e.g., Layer 0, 1):**
 - Tend to learn **local, surface-level linguistic features**.
 - They act like sophisticated **syntactic parsers**, focusing on dependencies between adjacent or nearby words (e.g., noun $\rightarrow$ verb, determiners $\rightarrow$ nouns, subject $\rightarrow$ predicate).
 - They establish **basic word sense** based on immediate context.
- **Middle Layers:**
 - Begin to **integrate global context** and resolve ambiguity.
 - They may focus on **semantic roles** (who did what to whom), **verb tenses**, and **scope of negation**.
 - Attention patterns start to become more diffused as they aggregate information from further back in the sequence.
- **Upper Layers:**
 - Focus on **high-level, abstract tasks** like **coreference resolution** (e.g., linking "Mr." to "Jaime Salazar" across long distances), **discourse structure**, and **overall sentence meaning/sentiment**.
 - They tend to have **sparse attention patterns** where a token attends very strongly to just one or two distant, critical tokens for the final prediction or representation.

Token Attention Shift

Yes, tokens attend to different tokens as we go through the layers.

1. **"dog"**: In Layer 0, "dog" attends primarily to "The". In a later layer, it might attend to the main verb of the *next* sentence to understand its role in the paragraph (e.g., if the next sentence is "It barked loudly," the later layer might have **"dog"** attending to **"It"** to resolve the anaphora).
2. **"running"**: In Layer 0, "running" attends to its immediate subject "Jaime". In a later layer, it might attend to a distant word like **"Mr."** to understand the **cohesive structure** and the overall topic being discussed (politics/elections) over multiple sentences.
3. **Coreference Tokens ("Mr.")**: While Layer 0 already establishes a strong link to "Jaime

yeah this explanation is way too long, I want to nudge it in the right direction

Salazar," subsequent layers will **reinforce this link** and aggregate the value vectors from the name tokens to create a **unified, context-aware representation** for "Mr."

Essentially, the layers progressively build abstract representations, starting with **syntax** and culminating in **semantics/discourse**.

Gentle nudge with the goal of ensuring that the model pays extra attentions to the images i feed it, instead of sidetracking into general explanations

User

Can you base your analysis only on what's happening in the three attention-head images I provided, instead of giving a general explanation of attention? I want you to describe what tokens are attending to what, and what similarities/differences you observe

Gemini

That is a great clarification. I will focus my analysis strictly on the three provided images for **Layer 0, Head 0**.

Focused Analysis of Layer 0, Head 0 Attention

Here is a token-by-token breakdown of the attention patterns observed in the three images:

1. Sentence: "The dog ran"

- **"The"** attends only to itself (a common pattern for the first token).
- **"dog"** attends most strongly to **"The"** and moderately to itself. This head is successfully linking the **article** to the **noun** it modifies.
- **"ran"** attends most strongly to **"dog"** and moderately to itself and **"The"**. This head is successfully identifying the **subject** for the **verb**.

2. Sentence: "The dog sitting in the car"

- **"The"** attends only to itself.
- **"dog"** attends most strongly to **"The"** and moderately to itself. (Article $\rightarrow$ Noun pattern is consistent).
- **"sitting"** attends most strongly to **"dog"**. This establishes the relationship between the **present participle** and its **subject**.
- **"in"** attends most strongly to **"dog"** and **"sitting"**. The **preposition** is linking itself to the preceding phrase structure.
- **"the"** attends most strongly to the preceding **"in"**.
- **"car"** attends most strongly to **"the"** and **"in"**. This head links the final **noun** to both the **article** and the **preposition** governing it.

3. Sentence: "Jaime Salazar is running for election. Despite his party affiliation, Mr."

- **Name Entity Cohesion:** The sub-word tokens of the name ("Ja", "ime", "Sal", "azar")

So this new response is shorter and more relevant to the images I have fed it, but still a bit too long. Will look to nudge the model again (so future questions are also short and succinct). In general the answers are pretty satisfactory

show a strong tendency to attend to each other, especially the **first tokens** ("Ja", "ime") attending to the **last tokens** ("Sal", "azar"). This suggests this head is helping to **cohere the sub-word tokens of a multi-token entity**.

- **Subject-Verb Link: "is" and "running"** attend strongly back to the name tokens ("Ja", "ime", "Sal", "azar").
- **Long-Range Coreference (Key Observation):** The final token, "Mr.", attends with remarkable strength and specificity to the tokens that form the name "**Jaime Salazar**" ("Ja", "ime", "Sal", "azar"). This demonstrates that **Layer 0, Head 0** is already capable of establishing **coreference links** across a punctuation mark and a dependent clause.

Similarities and Differences in Observation

Similarities

- **Local Syntax:** In all examples, the head shows a strong ability to capture **immediate, local syntactic relationships** in the same layer and head (Layer 0, Head 0). This includes Article $\rightarrow$ Noun ("The $\rightarrow$ dog") and Subject $\rightarrow$ Verb ("dog $\rightarrow$ ran" / "dog $\rightarrow$ sitting").
- **Self-Attention:** Tokens consistently show at least a moderate amount of attention to themselves.

Differences

Feature	Simple Sentences ("The dog ran," "The dog sitting...")	Complex Sentence ("Jaime Salazar...")
Attention Focus	The attention links are predominantly short-range (1-3 tokens back) and sparse (only 1-2 strong links per token).	The attention links are long-range and dense (e.g., "Mr." attends strongly across 10+ tokens).
Task Demonstrated	Primarily shows basic parts-of-speech dependencies (e.g., article modification, verb subject identification).	Demonstrates entity cohesion (sub-words attending to each other) and coreference resolution ("Mr." attending to "Jaime Salazar").
Attention to Sub-words	Not applicable; all tokens are single words.	Explicitly shows how attention is used to bind fragmented sub-word tokens into a single

		conceptual entity.
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User

Nudge #2: Be short and direct!

When I ask a question, please give a short and direct answer—just a few sentences, not long sections or detailed breakdowns. Focus only on what the assignment is asking for, based on the images or text I give you. Don't use markdown tables, long headers, or big explanations.

Gemini

Got it. I will keep my answers brief, direct, and focused strictly on the assignment's needs, avoiding detailed breakdowns, long headers, and markdown tables.



Next Question

User

Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?

Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.

Note: I have attached 5 different examples of layers

Gemini

Since the question involved attention graphs in which you could tweak the head and layer you were looking at, I made sure to provide Gemini with 5 examples that would simulate a "pass" through looking at available graph data

You're now looking at BERT, which uses **bidirectional** attention, unlike the **causal/unidirectional** attention in GPT.



Attention Patterns in BERT (Layers 2, 4, 6, 8)

- **Layer 2, Head 0 (Image 4):** Attention is very **local and sparse**, mainly showing tokens attending to their **immediate neighbors** (e.g., 'wanted' $\rightarrow$ 'I', 'have' $\rightarrow$ 'to'). It establishes basic **syntactic structure** within each sentence.
- **Layer 4, Head 0 (Image 6):** The pattern becomes slightly **denser** and begins to cross the separator. Crucially, the pronoun **'it'** in sentence B is starting to attend to the noun **'party'** in sentence A, establishing the **coreference link**.
- **Layer 6, Head 0 (Image 5):** The attention is now **highly distributed and dense**. Every token in sentence B attends to almost **every token in sentence A**, suggesting this layer is aggressively **integrating cross-sentence context** and forming a combined representation.
- **Layer 8, Head 0 (Image 3):** Attention becomes **sparse and specialized** again. The **[SEP]** tokens and the **[CLS]** token are heavily attended to by many tokens. This suggests

PART
B, i and
ii

a shift to **encoding high-level sentence/sequence meaning** into these special tokens for downstream tasks.

GPT vs. BERT Attention Differences

- **BERT (Bidirectional):** Tokens can attend to **all preceding and subsequent tokens** in the sequence. This enables better **contextual understanding** (e.g., 'I bought a cake for **it**' can attend to 'party' to resolve the pronoun).
- **GPT (Causal/Unidirectional):** Tokens can **only attend to preceding tokens**. This is required for **text generation**, as the model cannot "see" the future words it is about to predict.

User

Model does well to capture qualitative trends and distinguish BERT vs GPT. Matches the Official solution pretty well.

PART
B iii

Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?

Note: I attached 5 examples for sentence a and 5 corresponding layer examples for sentence b

Gemini included 5 examples for each graph, since gemini allows 10 images max

Yes, the attention patterns learned for the two sentences are demonstrably different, especially around the polysemous word **"play"**, which shows BERT's ability to **disambiguate word senses** based on context.

Contextual Disambiguation

The difference in attention shows that the embeddings for the tokens are being adjusted for the specific meaning:

- **Sentence A (Verb Sense: "I was going to play at the park..."):** The token **"play"** strongly attends to the **subject 'I'** and the **location 'park'** across multiple layers. This reinforces its role as a **verb** and an action taking place at a specific location. In later layers, the head often captures the **verb $\rightarrow$ subject** relationship, solidifying the action.
- **Sentence B (Noun Sense: "I was going to a play in the park..."):** The token **"play"** strongly attends to the **article 'a'** that precedes it in the early layers, and the **preposition 'in'** and the **location 'park'** in the middle layers. This reinforces its role as a **noun** or **object** that is modified by an article and governed by a preposition.
- **Result:** The unique attention weights applied in the respective sentences—looking at 'I' and 'at' (Sentence A) versus 'a' and 'in' (Sentence B)—result in two distinct **contextualized embeddings** for the token **"play"** in the final layers, effectively

solution is well aligned with the official solution, but i wanted it to be more specific about examples from the graphs

distinguishing the two senses.

User

can you find any examples in the attention graphs i provided?

Gemini

Yes, the attention graphs clearly show the **contextual disambiguation** of the word **"play"** between the two sentences.

- **Sentence A (Verb Sense: "I was going to play at the park..."):** In **Layer 6, Head 0** (Example A), the token **"play"** shows strong attention back to its **subject 'I'** and to the **preposition 'at'**. This highlights its function as an **action (verb)** whose query is focusing on the agent ('I') and its location cue ('at').
- **Sentence B (Noun Sense: "I was going to a play in the park..."):** In the **same layer and head** (Layer 6, Head 0, Example B), the token **"play"** shows strong attention to the **article 'a'** that precedes it and the **preposition 'in'**. This indicates its function as an **object/entity (noun)**, with its query focusing on the determiners ('a') and governing prepositions ('in') that modify a noun phrase.

This difference in the tokens attended to by the query of **"play"** in the same head across sentences is how BERT dynamically **adjusts the word's embedding** to represent its correct context-specific meaning.

User

pulls example from the graphs! happy with this update

PART B iv

Answer the following questions in your writeup:

Do you notice BERT's bidirectionality in play?

Do you think pre-training the BERT helped it learn better representations?

I provided 5 layer examples from the first graph and 5 layer examples from the second. Please include these when reasoning.

Gemini

Bidirectionality and Pre-training in BERT

Do you notice BERT's bidirectionality in play?

Yes, BERT's bidirectionality is clearly visible in the attention graphs. The most important example is the **contextual disambiguation** of the word **"play"**. The token is able to look **forward** at context cues like the **preposition 'in'** (Sentence B) or **'at'** (Sentence A) and use those **future cues** to adjust its current representation, which is a feature of bidirectionality. Additionally, in the longer examples, tokens attend to words both before and after the [SEP] token, demonstrating full access to all context.

Do you think pre-training the BERT helped it learn better representations?

Absolutely, yes. The visualizations show clear evidence of pre-training effectiveness. The model develops layers and heads specialized for **sophisticated linguistic tasks**. For example, the difference in attention patterns for the ambiguous word **'play'** between Sentence A and Sentence B, even in the same head (Layer 6, Head 0), proves that pre-training enables the model to create two distinct, superior **contextualized embeddings** for the same token based on its surrounding context.

User correct and matches official solution

PART C

Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Can you identify any of the different features that the different attention heads are focusing on?

Gemini

Yes, different features are clearly being learned across the different attention heads of BERT.

This is by design, as the **Multi-Head Attention** mechanism allows each head to **learn a unique set of projection matrices** (Query, Key, Value). This enables each head to focus on a **different type of linguistic relationship** simultaneously, providing the model with a richer, parallel perspective on the input.

Identified Features:

- **Syntax/Local Features (Early Heads):** Heads in early layers (e.g., Layer 2, Head 0) exhibit **sparse, local attention**, focusing on basic **article** $\rightarrow$ **noun** or **verb** $\rightarrow$ **subject** dependencies.
- **Coreference/Contextual Features (Middle Heads):** Heads like Layer 4, Head 0 specialize in **cross-sentence coreference**, linking pronouns (like 'it') to their antecedents (like 'party') across the sequence.
- **Structural/Boundary Features (Late Heads):** Other heads in later layers (e.g., Layer 8, Head 0) focus strongly on the **special tokens** like **[CLS]** and **[SEP]**, acting as summarization heads that aggregate global context into these markers.

User correct and matches the official solution

What is the difference between a head and a layer

Gemini curious for some extra context

A **layer** refers to a **full computational block** in the transformer's sequential structure, containing a Multi-Head Attention (MHA) module and a Feed-Forward Network (FFN). Data

passes through these layers sequentially to refine the token representations.

A **head** is a single **independent instance** of the attention mechanism *within* the Multi-Head Attention module of a single layer. Each head operates in **parallel** with other heads in the same layer, learning a distinct set of weight matrices to focus on a unique linguistic aspect.

User

PART D

help me answer these please

Gemini



Attention Head Specialization in BERT

Yes, different features are clearly being learned across the different attention heads of BERT.

This happens because the **Multi-Head Attention** mechanism allows each head to **learn a unique set of Query, Key, and Value projection matrices**. This design permits each head to focus on a **different type of linguistic relationship** simultaneously, providing the model with a richer, parallel perspective on the input.

Identified Features:

correct, matches official solution

- **Syntax/Local Features (Early Heads):** Some heads in early layers (e.g., Layer 2, Head 0) show **sparse, local attention**, focusing on immediate neighbors like **article** $\rightarrow$ **noun** or **verb** $\rightarrow$ **subject**.
- **Coreference/Contextual Features (Middle Heads):** Heads like Layer 4, Head 0 specialize in **cross-sentence coreference**, such as linking the pronoun **'it'** in the second sentence to the noun **'party'** in the first sentence.
- **Structural/Boundary Features (Late Heads):** Other heads in later layers (e.g., Layer 8, Head 0) focus strongly on the **special tokens** like **[CLS]** and **[SEP]**, acting as summarization heads that aggregate global context into these markers.

User

PART E

Were you able to identify interesting patterns in the visualizations? If yes, please share some examples (describe in text or paste a screenshot). If not, feel free to use this space for your frustrations.

Gemini

Yes, I was able to identify several interesting patterns in the BERT visualizations, particularly related to the model's bidirectionality and its ability to specialize attention heads.



Examples of Interesting Patterns

1. **Coreference Resolution (Layer 4, Head 0, Sentence A/B):**

- The most striking pattern is the **long-range coreference** where the pronoun **'it'** in the second sentence starts to attend strongly to its antecedent, the word **'party'** (or similar noun) in the first sentence. This shows a head specializing in **semantic linking** across sentence boundaries.

2. **Word Sense Disambiguation (Layer 6, Head 0, Sentence A vs. B):**

- This head shows a clear difference in the tokens attended to by the polysemous word **"play"**:
 - **Sentence A (Verb Sense):** **"play"** attends strongly to the **subject 'I'** and the preposition **'at'**, reinforcing its function as an action (verb).
 - **Sentence B (Noun Sense):** **"play"** attends strongly to the **article 'a'** and the preposition **'in'**, reinforcing its function as an object (noun).

3. **Boundary and Summarization Attention (Later Layers, e.g., Layer 8, Head 0):**

- In the deeper layers, many tokens put a high percentage of their attention mass onto the **special tokens** like **[CLS]** and **[SEP]**. This indicates the head is specializing in **global context aggregation**, funneling the entire sentence's meaning into the **[CLS]** token for classification or next-sentence prediction.

These distinct patterns in different heads and layers confirm that BERT learns a rich set of specialized linguistic features from its pre-training.